

Muhammad Mikaal Amjad

Full Stack Engineer • AI & LLM Integration

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Lahore, Pakistan

PROFILE

Full Stack Engineer with 2+ years of professional experience building AI-powered, data-intensive web applications using Python, TypeScript, Next.js 15, FastAPI, and Django REST Framework. Specialise in LLM integration, RAG pipelines, and agentic workflows — combining graph databases (Neo4j), vector retrieval, and OpenRouter LLM APIs to deliver intelligent production features. Proven across the full stack: event-driven backend architectures, real-time WebSocket systems, ETL pipelines, and modern TypeScript frontends following Atomic Design. Deliver end-to-end — from schema design to cloud deployment on Azure and AWS.

TECHNICAL SKILLS

AI & LLM: OpenRouter LLM, Claude API, RAG Pipelines, Agentic Workflows, MCP Protocols, Prompt Engineering, LLM Streaming

Graph & Vector: Neo4j (Certified), Graph-backed RAG, Cypher, Knowledge Graphs

Backend: FastAPI, Django REST Framework, Node.js, SQLAlchemy, Alembic, Celery, Redis (Pub/Sub & Producer/Consumer), WebSockets

Frontend: Next.js 15, React, TypeScript, JavaScript, Atomic Design

Databases: PostgreSQL, MS SQL, MongoDB, Neo4j, Redis

Cloud & DevOps: Azure (Blob, Data Factory, ACS, Event Grid, Entra ID), AWS (Textract, S3), Docker, Coolify, CI/CD, Git

Pipelines & Data: Apache Airflow, ETL, Azure Data Factory, Pandas, Geospatial Ingestion

Integrations: Microsoft Graph API, Google Drive API, Google Maps API, AWS Textract, Azure Communication Services, Webhooks

CERTIFICATIONS

[Neo4j Certified Professional](#) • Neo4j

[Microsoft Certified](#) • Microsoft

[DataCamp Certified](#) • DataCamp

PROFESSIONAL EXPERIENCE

Software Engineer | Devsinc

April 2025 – Present • Lahore, Pakistan (On-site)

- Built RAG pipelines combining Neo4j graph database with OpenRouter LLM for AI-assisted knowledge retrieval across large enterprise datasets — enabling semantic search over structured graph data.
- Implemented OpenRouter LLM and agentic workflows for AI-assisted production features — agent planning, multi-step tool use, and fallback handling in live environments.
- Designed event-driven document ingestion pipeline — Azure Event Grid triggers Airflow DAG on new Blob Storage files, automatically processing and storing structured data end to end.
- Integrated AWS Textract for automated PDF and document OCR extraction — pulling structured data from real documents into the pipeline without manual intervention.
- Implemented Redis for inter-service communication using Producer/Consumer and Pub/Sub patterns for reliable async microservice architecture with failure isolation.
- Built real-time WebSocket server for live data delivery; integrated push, email, and in-app notification services for user-facing workflows.
- Built FastAPI backends with Neo4j, PostgreSQL, MS SQL, and MongoDB; developed Next.js 15 TypeScript frontends using Atomic Design methodology.
- Engineered ETL pipelines using Apache Airflow and Azure Data Factory for multi-source geospatial data ingestion, plotting origin coordinates via Google Maps API.

- Built Azure Communication Services email workflows with webhook tracking — identifying reply threads by ref# and updating the database in real time.

Freelance Software Engineer | Self-Employed

2023 – 2026 • Remote

- Built fleet management portal for Masar Pty Ltd, Australia (manage.masarptyltd.com) — automated vehicle lifecycle, driver/rental management, document handling, and photo uploads via Google Workspace Shared Drive API. 16-table PostgreSQL schema, built solo, live in production.
- Built Shield Cyber Services CRM and diagnostic platform (shieldcyberservices.com) — automated cyber risk scoring engine, client management pipeline, and diagnostic workflows for UK SMEs with full Microsoft 365 integration.
- Developed an AI chatbot integrating LLM APIs with prompt engineering for automated client-facing conversational workflows.
- Deployed all platforms on OVHcloud dedicated servers using Coolify for Docker-based container orchestration.
- Built the Zoomautos.co.uk web platform (full-stack), integrating vehicle listing management and user-facing search.

PROJECTS

NCAA Football Analytics Platform — Python · FastAPI · Next.js 15 · TypeScript · Claude API · Docker

- AI-powered analytics platform using Claude API for game outcome prediction and performance analysis across college football datasets.
- Built FastAPI backend with structured data ingestion, Next.js 15 TypeScript frontend, and Dockerised deployment pipeline.

github.com/MmikaalAmjad/football-suggestion

NutriSuggest — AI Meal Suggestion App — Python · FastAPI · Next.js · RAG · LLM Integration

- RAG-powered meal recommendation engine — ingests nutritional data, indexes it for retrieval, and serves personalised suggestions via LLM.
- End-to-end: FastAPI backend, retrieval pipeline, and Next.js frontend with real-time LLM streaming responses.

github.com/MmikaalAmjad/meal-suggestion

GeoSphere — Geospatial Web Application — Next.js · Python · Neo4j · Google Maps API · Apache Airflow

- Geospatial data platform ingesting multi-source location datasets via Airflow DAGs, storing in Neo4j for graph-based spatial queries.
- Next.js frontend with Google Maps API integration for interactive coordinate visualisation.

github.com/MmikaalAmjad/geosphere

Financial Data Ingestion Backend — Python · FastAPI · Apache Airflow · PostgreSQL · ETL · Pandas

- Multi-source financial ETL backend — Airflow-orchestrated DAGs ingest, clean, and load structured financial data into PostgreSQL.
- FastAPI layer exposes processed data for downstream consumption; Pandas used for transformation and anomaly handling.

github.com/MmikaalAmjad/financial-data-ingestion-backend

Masar Pty Ltd — Fleet Management Portal — Next.js · Python · PostgreSQL · Google Drive API · Docker

- Full-stack fleet management system for an Australian client — vehicle lifecycle automation, driver/rental workflows, document storage via Google Workspace Shared Drive API.
- 16-table PostgreSQL schema, Dockerised, built and shipped solo. Live in production at manage.masarptyltd.com.

manage.masarptyltd.com

EDUCATION

Bachelor of Software Engineering

COMSATS University Islamabad — Lahore Campus • Graduated 2025